scTAM-seq enables targeted high-confidence analysis of DNA methylation in single cells

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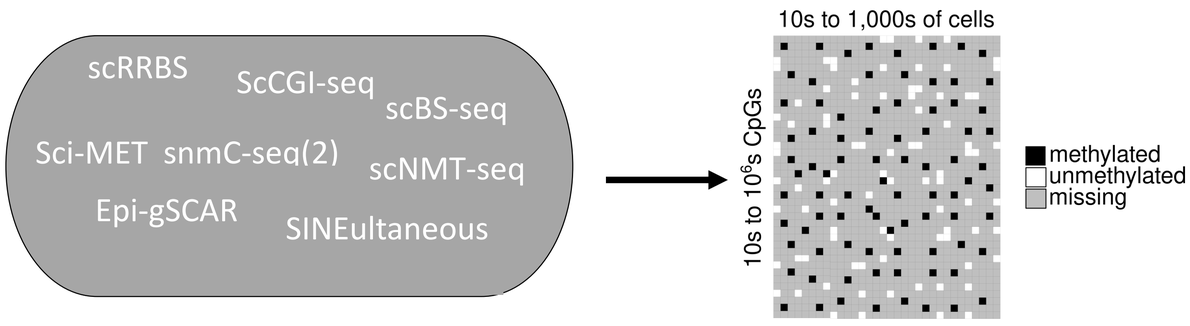
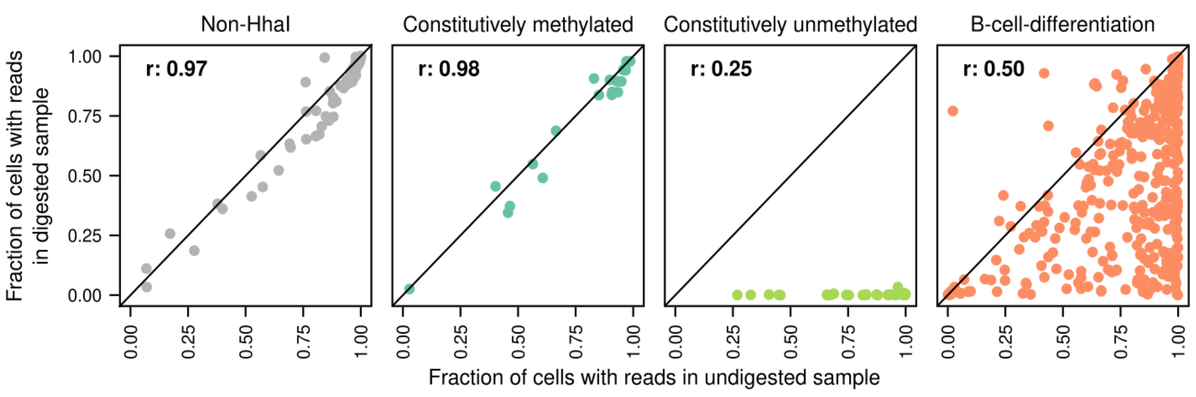
Genome-wide single-cell DNA methylation methods generate extremely sparse data matrices



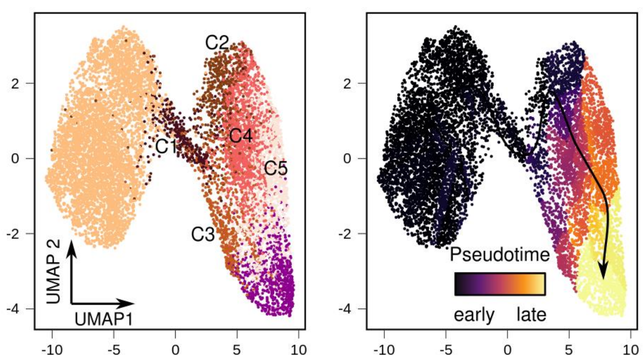
- Many genome-wide, single-cell DNA methylation (**scDNAm**) methods have been developed
- Methods require **bisulfite conversion** and **prohibitive sequencing effort**
- The result is an **extremely sparse** data matrix

Technical controls show low false positive and false negative rates and allow for selecting high-confidence amplicons

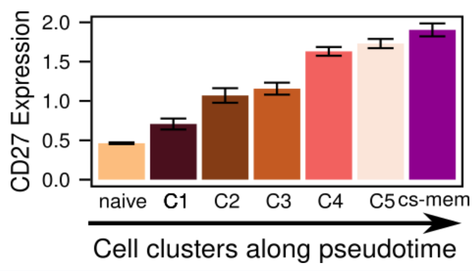


- **False Negative Rate** of as low as **7%** and **False Positive Rate** less than **0.2%**

scTAM-seq can be used to characterize the DNA methylation profiles of peripheral blood B-cell subtypes

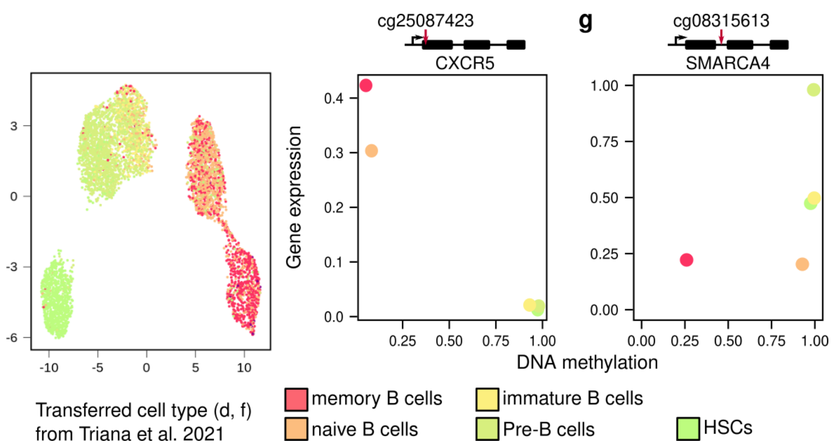


- **Standard** scRNA-seq software workflows (Seurat[1], Signac[2]) can be used

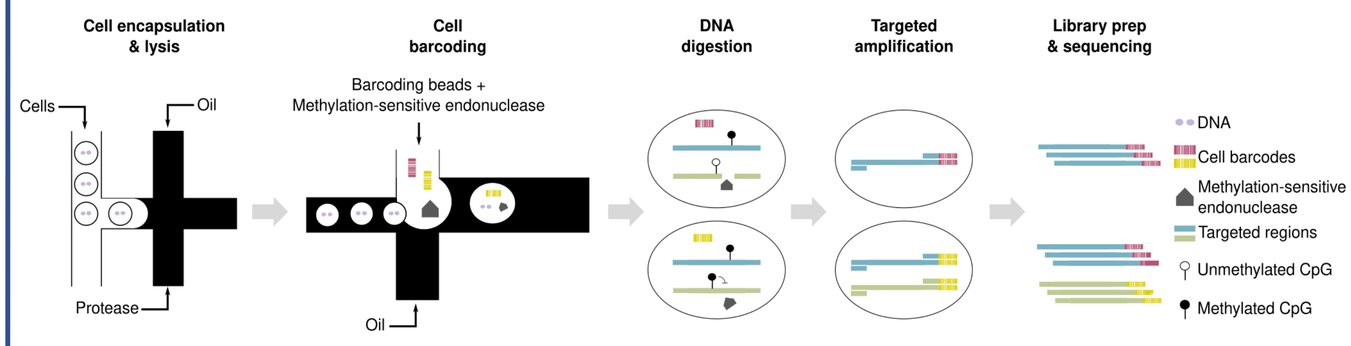


Surface marker expression allows for integration with single-cell proteo-transcriptomic atlasses[3]

- Data integration was conducted using the surface marker expression as **anchors**
- DNA methylation can be associated with **changes in gene expression** on the cell-type level

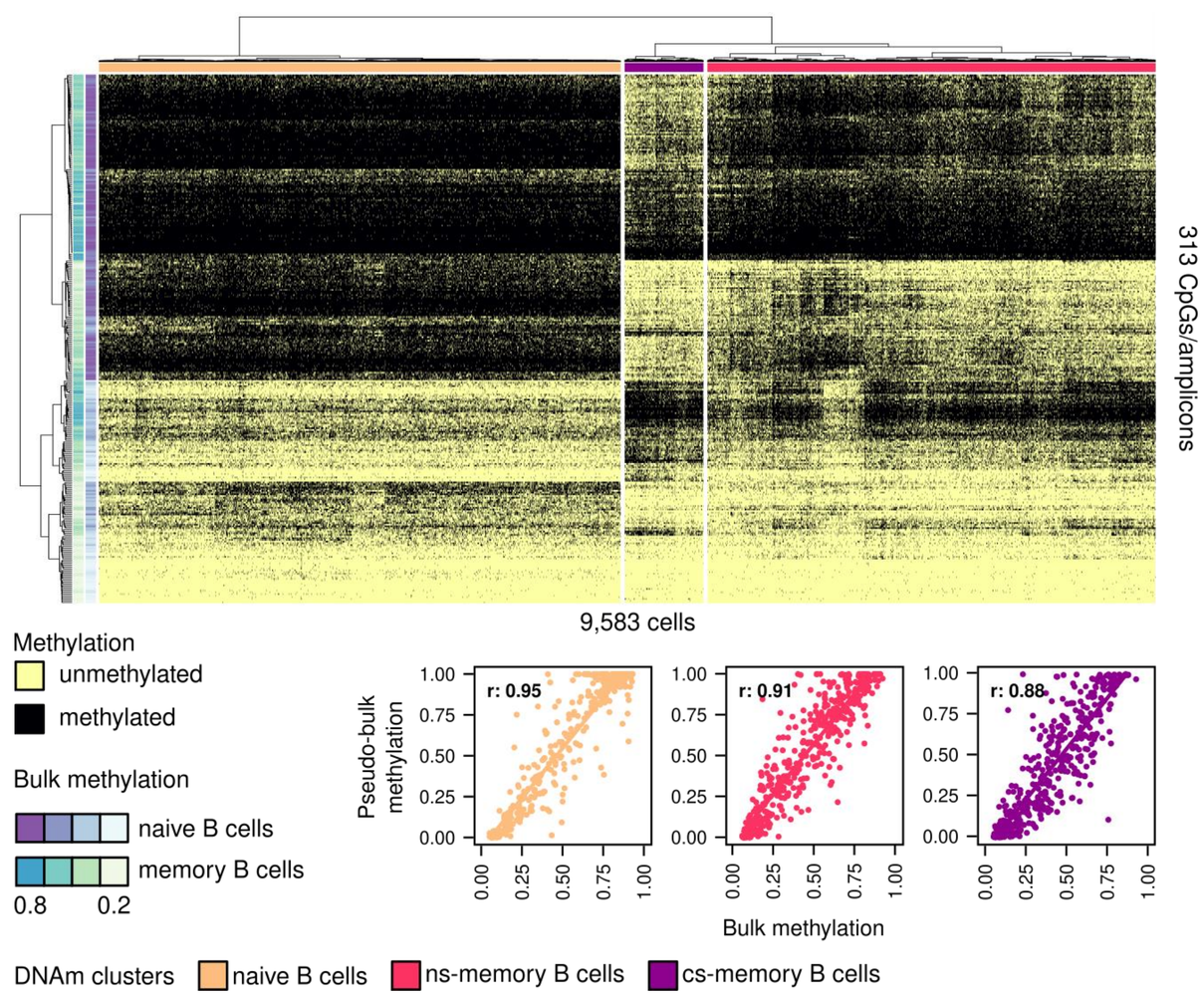


scTAM-seq allows for investigating DNA methylation of up to 1,000 CpGs in up to 10,000 cells



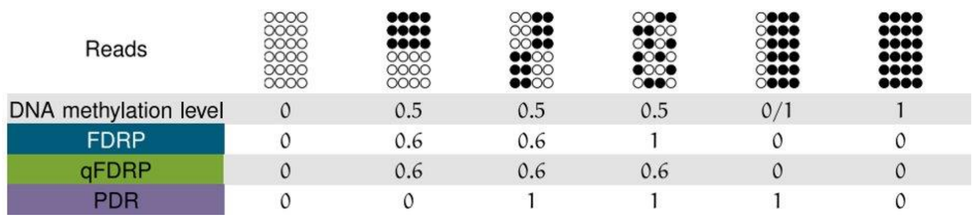
- scTAM-seq is a **targeted, bisulfite-free** scDNAm method
- scTAM-seq utilizes the **Mission Bio Tapestri** technology making it **widely applicable**
- Platform allows joint analysis of DNA methylation with **surface marker expression** and **somatic mutations**

scDNAm data can be used to classify cells according to their cell types



- Bulk information is used to assign clusters to cell types
- **Bayesian inference** to estimate pseudo-bulk methylation values, which strongly correlate with available bulk data
- Cell type assignment is corroborated by surface marker expression

Bulk data can be leveraged for selecting sites variably methylated in a cell population of interest



From Scherer et al., NAR, 2020[4]

- **Bulk** bisulfte sequencing data comprises DNA methylation information at the **single-molecule level**
- Variably methylated sites can be selected for **in-depth** profiling with scTAM-seq
- We developed a **pipeline** automating the **selection of CpGs** for scTAM-seq from bulk data (https://github.com/veltenlab/CpGSelectionPipeline)

Preprint

Bianchi and Scherer, et al., *bioRxiv*, 2022, doi: 10.1101/2022.04.11.487648

R-resources: https://github.com/veltenlab/scTAM-seq-scripts
https://github.com/veltenlab/CpGSelectionPipeline

GitHub

References

[1]Satija, R., et al., *Nat Biotechnol*, 2015, doi: 10.1038/nbt.3192
[2]Stuart, T., et al., *Nat Methods*, 2021, doi: 10.1038/s41592-021-01282-5
[3]Triana, S., et al., *Nat Immunol*, 2021, doi: 10.1038/s41590-021-01059-0
[4]Scherer, M., et al., *NAR*, 2020, doi: 10.1093/nar/gkaa120

CONCLUSIONS

- ScTAM-seq enables high-confidence analysis of DNA methylation for up to **1,000 CpGs** in up to **10,000 single cells**
- DNA methylation can be used to resolve novel cellular states previously hidden in bulk data
- Bulk data can be leveraged for selecting CpGs for scTAM-seq

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